Conservation of Energy Problems

Answer key

Conventions used throughout

• **Energy symbols (from the lessons):** W = work, E_k = kinetic energy ($\frac{1}{2}mv^2$), E_g = gravitational potential energy (mgh), E_e = elastic potential energy ($\frac{1}{2}kx^2$). x is the stretch or compression measured from the spring's unstretched length.

• **Method (from the lesson):** draw the situation, list the kinds of energy at the start and at the end, then write $E_{TOTi} = E_{TOTf}$. With friction or another non-conservative force: $E_{TOTi} - W_F = E_{TOTf}$, where W_F is the size of the work done by that force.

• **Heights:** E_g is relative, so each problem picks its own h = 0 (usually the lowest point of the object).

• **Friction:** $F_f = \mu F_N$, and the work it removes is $W_F = F_f \Delta d$ over the whole distance slid. The normal force comes from the balanced direction ($F_{net,y} = 0$).

• **Connected objects:** the string does not stretch, so both objects move the same distance and have the same speed (question 20 is the exception, and says why).

• **Quadratics:** when the unknown appears squared and to the first power, the key uses the quadratic formula and keeps the root that fits the situation.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** work and energy are scalars (a size and a sign, no direction). Vectors are given as a size with the direction in square brackets.

• g = 9.81 m/s^2. Every step keeps full precision; only final answers are rounded.

QUESTION 1

Freight elevator onto a spring

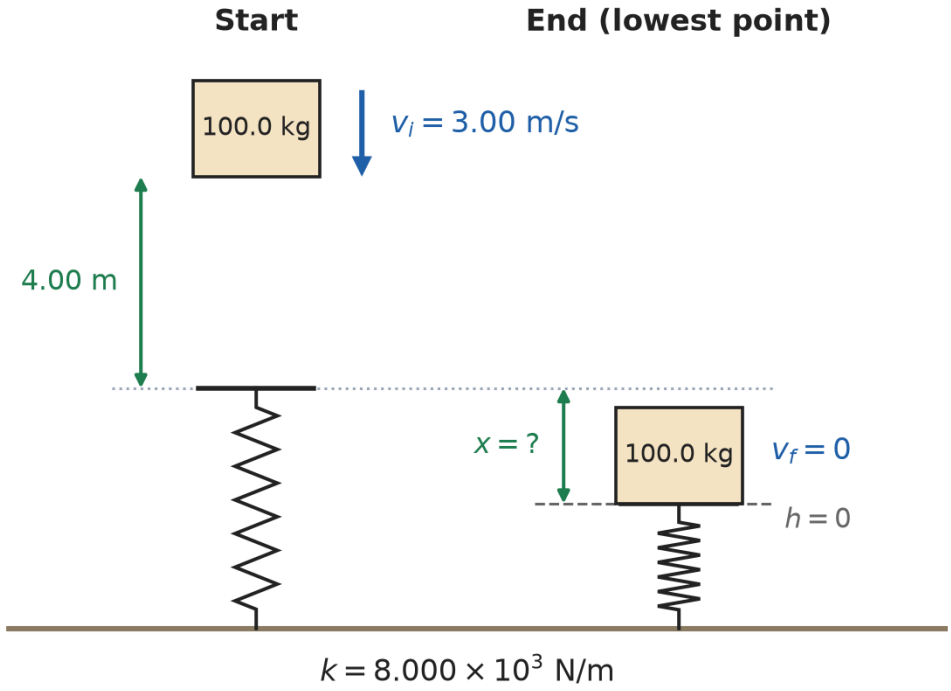
1. A freight elevator with a total mass of 100.0 kg is moving downward at 3.00 m/s when the cable snaps. The car falls 4.00 m onto a huge spring with a spring constant of 8.000 x 10^3 N/m. By how much will the spring be compressed when the elevator stops moving?

GIVEN

$m = 100.0$ kg $\quad v_i = 3.00$ m/s $\quad \Delta d = 4.00$ m

$k = 8.000 \times 10^3$ N/m $\quad x = ?$

The elevator when the cable snaps and at its lowest point



h = 0 is the lowest point, so the elevator starts (4.00 + x) above it.

Step 1 – Energy equation

Start: the elevator is moving and is (4.00 + x) above h = 0, so it has E_k and E_g. End: it is at rest at h = 0 on the compressed spring, so all the energy is E_e.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mg(\Delta d + x) + \frac{1}{2}mv_i^2 = \frac{1}{2}kx^2$

Move everything to one side:

$$\frac{1}{2}kx^2 - mgx - \left(mg\,\Delta d + \frac{1}{2}mv_i^2\right) = 0$$

Put in the numbers: $4000x^2 - 981.0x - 4374.0 = 0$

Here ½k = 4000, mg = (100.0)(9.81) = 981.0, and mgΔd + ½mv_i^2 = 3924.0 + 450.0 = 4374.0.

Step 2 – Solve the quadratic

Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put in the numbers:

$$x = \frac{-(-981.0) \pm \sqrt{(-981.0)^2 - 4(4000)(-4374.0)}}{2(4000)}$$

$$x = \frac{981.0 \pm 8422.97}{8000}$$

$x = 1.1755$ m or $x = -0.9302$ m

A compression cannot be negative, so x = 1.1755 m.

FINAL ANSWER

x = 1.18 m

QUESTION 2

Elevator cable breaks

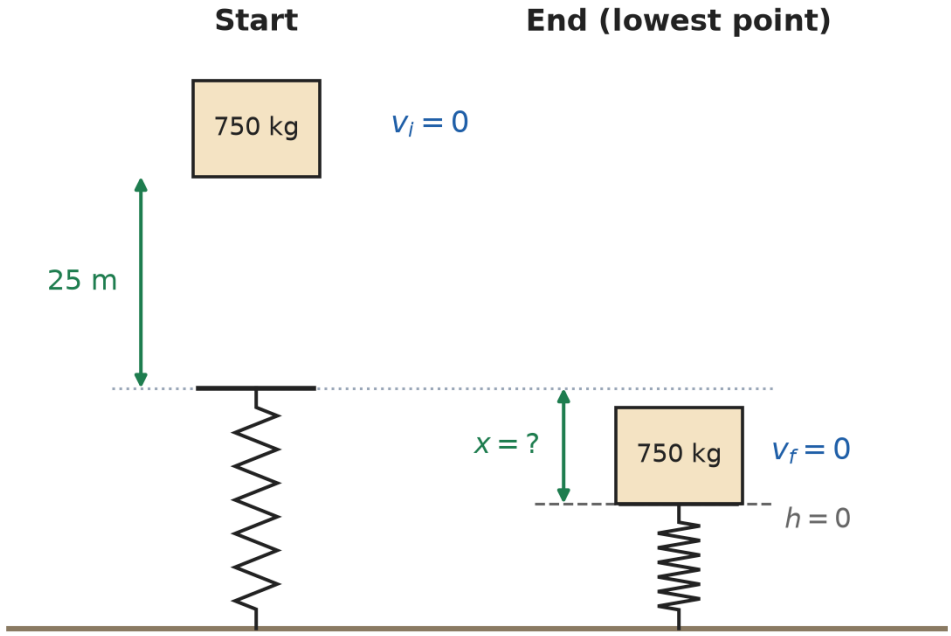
2. An elevator cable breaks when a 750 kg elevator is 25 m above a huge spring (k = 4.0 x 10^4 N/m) at the bottom of the shaft. Calculate (a) the work done by gravity on the elevator just before it hits the spring and b) the speed of the elevator just before striking the spring (c) the amount the spring compresses.

GIVEN

$m = 750$ kg $\quad v_i = 0 \quad \Delta d = 25$ m

$k = 4.0 \times 10^4$ N/m $\quad x = ?$

The elevator when the cable breaks and at its lowest point



$k = 4.0 \times 10^4$ N/m

(a) the work done by gravity on the elevator just before it hits the spring

Gravity points down and the displacement is down, so the angle between them is 0° and the work is positive.

Formula: $W = (F\cos\theta)\,\Delta d$

Put in $F = mg$:

$$W_g = (mg\cos 0°)\,\Delta d = (750)(9.81)(1)(25) = 183937.5 \text{ J}$$

b) the speed of the elevator just before striking the spring

The elevator starts from rest, and gravity is the only force doing work, so the work done by gravity equals its kinetic energy at the spring. The mass cancels.

Formula: $W_g = \Delta E_k \quad \Rightarrow \quad mg\,\Delta d = \frac{1}{2}mv^2$

Divide by m, multiply by 2, square root: $v = \sqrt{2g\,\Delta d}$

Put in the numbers: $v = \sqrt{2(9.81)(25)} = 22.147$ m/s

(c) the amount the spring compresses

Step 1 – Energy equation

Take h = 0 at the lowest point. The elevator starts at rest (25 + x) above it and ends at rest on the compressed spring.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mg(\Delta d + x) = \frac{1}{2}kx^2$

Move everything to one side: $\frac{1}{2}kx^2 - mgx - mg\,\Delta d = 0$

Put in the numbers: $20000x^2 - 7357.5x - 183937.5 = 0$

Step 2 – Solve the quadratic

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put in the numbers:

$$x = \frac{-(-7357.5) \pm \sqrt{(-7357.5)^2 - 4(20000)(-183937.5)}}{2(20000)}$$

$$x = \frac{7357.5 \pm 121528.32}{40000}$$

$x = 3.2221$ m or $x = -2.8543$ m

A compression cannot be negative, so x = 3.2221 m.

FINAL ANSWER

(a) $W_g = 1.8 \times 10^5$ J

b) v = 22 m/s

(c) x = 3.2 m

QUESTION 3

Mass released on a vertical spring

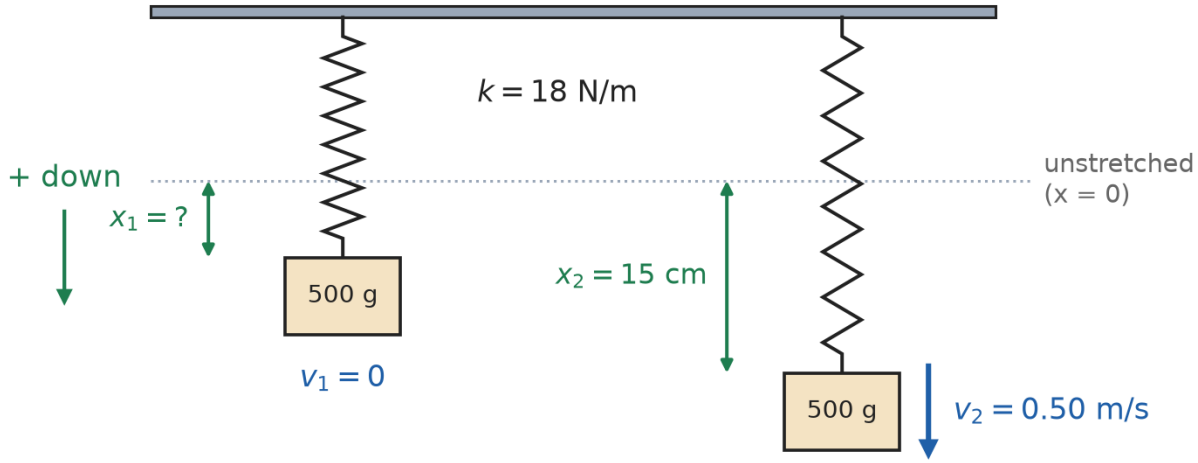
How the question is read: “equilibrium” is taken as the spring’s unstretched position (x = 0), the way Lesson 03 defines it. The worksheet’s answer is also measured from the unstretched position.

3. A 500 g (3SD) mass is released from rest from some distance below its equilibrium position and allowed to fall freely. If its velocity 15 cm below equilibrium is 0.50 m/s and the spring constant is 18 N/m, from what position was it released?

GIVEN

$m = 0.500$ kg	$v_1 = 0$
$x_2 = 15$ cm $= 0.15$ m	$v_2 = 0.50$ m/s
$k = 18$ N/m	$x_1 = ?$

The mass when it is released and 15 cm below the unstretched position



Released (at rest) **15 cm below**

x is the stretch of the spring, measured down from its unstretched position. h = 0 is the 15 cm point.

Step 1 – Energy equation

Start: at rest, with the spring stretched x_1, and $(x_2 - x_1)$ above h = 0. At the 15 cm point: moving, with the spring stretched x_2.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}kx_1^2 + mg(x_2 - x_1) = \frac{1}{2}kx_2^2 + \frac{1}{2}mv_2^2$

Move everything to one side:

$$\frac{1}{2}kx_1^2 - mgx_1 + \left(mgx_2 - \frac{1}{2}kx_2^2 - \frac{1}{2}mv_2^2\right) = 0$$

Put in the numbers: $9.0x_1^2 - 4.905x_1 + 0.47075 = 0$

Here mg = (0.500)(9.81) = 4.905, and the bracket is 0.73575 − 0.2025 − 0.0625 = 0.47075.

Step 2 – Solve the quadratic

Quadratic formula: $x_1 = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put in the numbers:

$$x_1 = \frac{-(-4.905) \pm \sqrt{(-4.905)^2 - 4(9.0)(0.47075)}}{2(9.0)}$$

$$x_1 = \frac{4.905 \pm 2.6668}{18.0}$$

$x_1 = 0.4207$ m or $x_1 = 0.1243$ m

Both answers are positions where the mass is at rest: the top and the bottom of its bounce. It was released and then fell to the 15 cm point, so it started above that point: x_1 = 0.1243 m.

FINAL ANSWER

Released 0.12 m below the unstretched position

QUESTION 4

Toy dart

4. A child's toy shoots a rubber dart horizontally using a compressed spring with a force constant of 3.5 x 10^2 N/m. The dart has a mass of mass 7.8 g and the spring is initially compressed 4.5 cm.

a. What is the elastic potential energy of the spring?

b. What is the speed with which the dart leaves the toy?

GIVEN

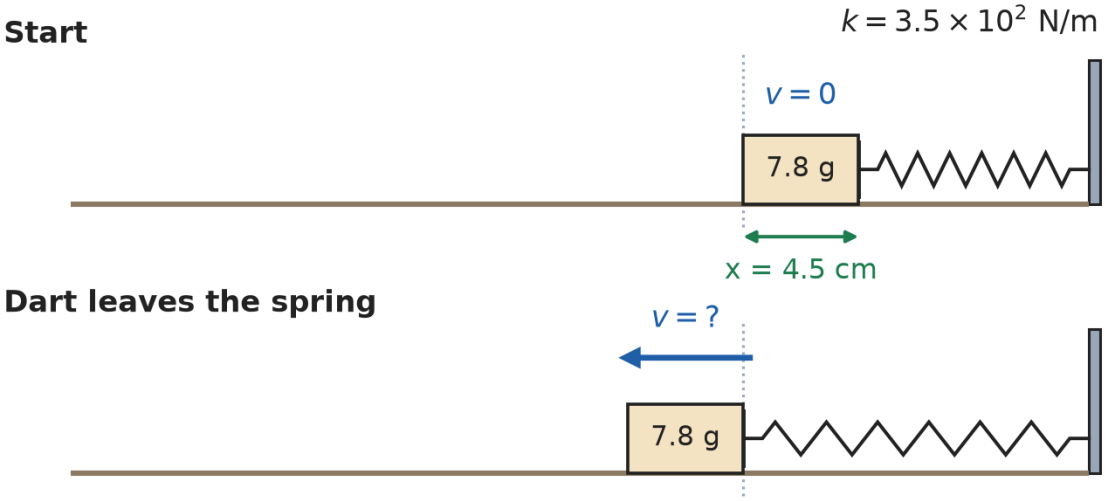
$k = 3.5 \times 10^2$ N/m

$m = 7.8$ g $= 0.0078$ kg

$x = 4.5$ cm $= 0.045$ m

$v = ?$

The dart before and after the spring is released



a. What is the elastic potential energy of the spring?

Formula: $E_e = \frac{1}{2}kx^2$

Put in the numbers: $E_e = \frac{1}{2}(3.5 \times 10^2)(0.045)^2 = 0.35438$ J

b. What is the speed with which the dart leaves the toy?

The dart moves horizontally, so E_g does not change. All of the elastic energy becomes the dart's kinetic energy.

Formula: $E_{TOTi} = E_{TOTf} \quad \Rightarrow \quad E_e = \frac{1}{2}mv^2$

Multiply by $\frac{2}{m}$ and square root: $v = \sqrt{\frac{2E_e}{m}}$

Put in the numbers: $v = \sqrt{\frac{2(0.35438)}{0.0078}} = 9.532 \text{ m/s}$

FINAL ANSWER

a. E_e = 0.35 J

b. v = 9.5 m/s

QUESTION 5

Block fired up a ramp

5. In a game, a small block is fired from a compressed spring up a plastic ramp into various holes for scoring. The mass of the block is 3.5 x 10^{-3} kg. The spring's force constant is 9.5 N/m. Friction is negligible.

a. If the block is to slide up the ramp through a vertical height of 5.7 cm, by how much must the spring be compressed?

b. If friction were not negligible, would your answer in (a) increase, decrease or stay the same? Explain.

GIVEN

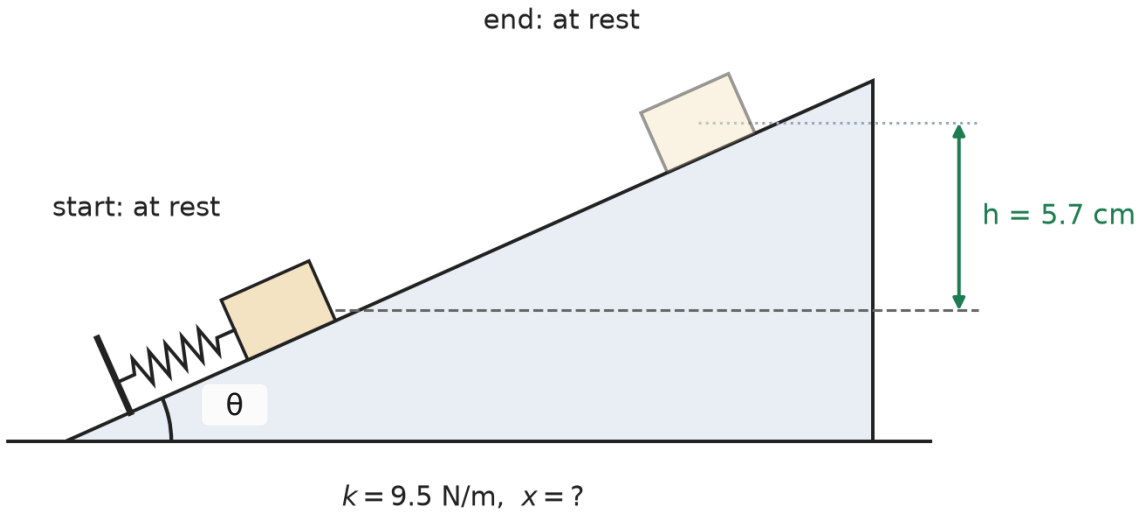
$m = 3.5 \times 10^{-3}$ kg

$k = 9.5$ N/m

$h = 5.7$ cm $= 0.057$ m

$x = ?$

The block at the start and at the top of its slide (ramp angle not given)



h = 0 is the starting position. The ramp angle is not needed: only the vertical height matters.

a. If the block is to slide up the ramp through a vertical height of 5.7 cm, by how much must the spring be compressed?

Start: at rest on the compressed spring, so the energy is all elastic. End: just reaching the height h, at rest, so it is all gravitational.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}kx^2 = mgh$

Multiply by $\frac{2}{k}$ and square root: $x = \sqrt{\frac{2mgh}{k}}$

Put in the numbers:

$$x = \sqrt{\frac{2(3.5 \times 10^{-3})(9.81)(0.057)}{9.5}} = 0.02030 \text{ m}$$

b. If friction were not negligible, would your answer in (a) increase, decrease or stay the same? Explain.

Increase. Friction removes some of the energy on the way up ($E_{TOTi} - W_F = E_{TOTf}$), so the spring must store more energy at the start to still give the block mgh at the top. More stored energy means a larger compression.

FINAL ANSWER

a. x = 0.020 m = 2.0 cm

b. Increase: the spring must also supply the energy that friction removes

QUESTION 6

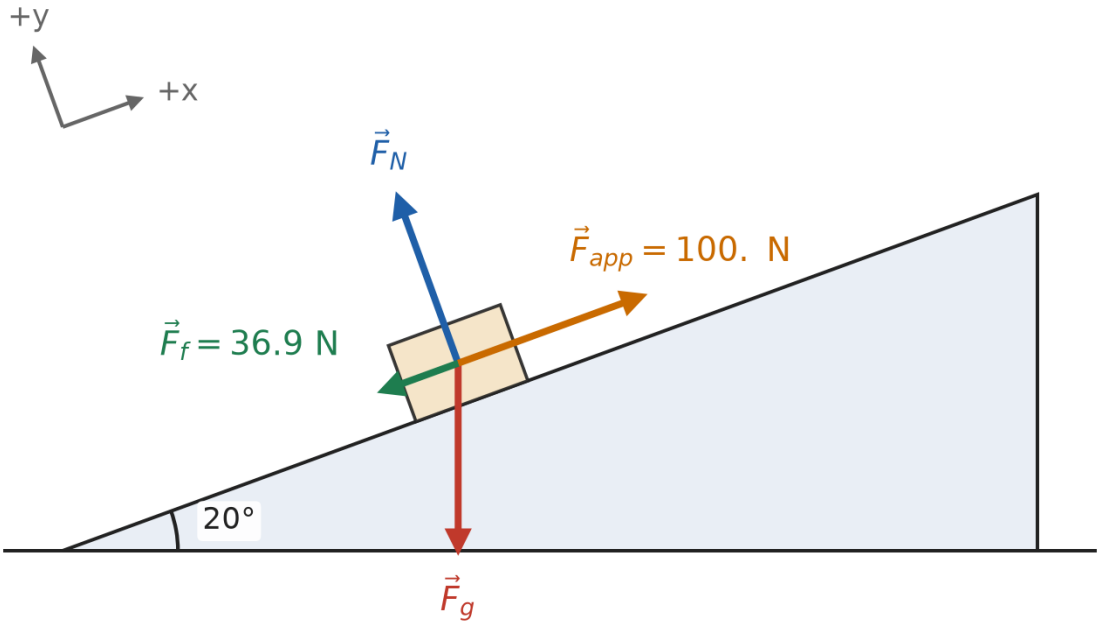
Crate pulled up a rough incline

6. A crate of mass 10.0 kg is pulled up a rough incline with an initial speed of 1.50 m/s. The pulling force is 100. N parallel to the incline and the incline makes an angle of 20.0^{o} to the horizontal. The force of friction between the crate and the ramp is 36.9 N and the crate is pulled for 5.00 m. (a) How much work is done by gravity? (b) How much energy is lost due to friction? (c) How much work is done by the 100 N force? (d) What is the speed of the crate after being pulled up the 5.00 m?

GIVEN

$m = 10.0$ kg	$v_1 = 1.50$ m/s	$F_{app} = 100.$ N
$\theta = 20.0°$	$F_f = 36.9$ N	$\Delta d = 5.00$ m

FBD of the crate (+x up the incline, the direction of motion)



(a) How much work is done by gravity?

The crate rises a height h = Δd sin θ. Gravity points down while the crate moves up, so gravity does negative work, equal to minus the gain in E_g.

Height gained: $h = \Delta d \sin\theta = (5.00)\sin 20.0° = 1.7101\ \text{m}$

Formula: $W_g = -mgh$

Put in the numbers: $W_g = -(10.0)(9.81)(1.7101) = -167.76\ \text{J}$

(b) How much energy is lost due to friction?

Friction acts along the whole 5.00 m, against the motion. The energy lost is the size of the work it does.

Formula: $W_F = F_f \Delta d$

Put in the numbers: $W_F = (36.9)(5.00) = 184.5\ \text{J}$

(c) How much work is done by the 100 N force?

The pull is parallel to the incline, in the direction of motion, so θ = 0° between the force and the displacement.

Formula: $W = (F\cos\theta)\,\Delta d$

Put in the numbers: $W_{app} = (100.)(\cos 0°)(5.00) = 500.0\ \text{J}$

(d) What is the speed of the crate after being pulled up the 5.00 m?

Take h = 0 at the start. The pulling force adds energy to the crate and friction removes it. At the end the crate has kinetic energy and gravitational energy mgh (the 167.76 J from part (a)).

Formula: $E_{TOTi} + W_{app} - W_F = E_{TOTf}$

Put in the energies: $\frac{1}{2}mv_1^2 + W_{app} - W_F = \frac{1}{2}mv_2^2 + mgh$

Isolate v_2: $v_2 = \sqrt{\frac{2\left(\frac{1}{2}mv_1^2 + W_{app} - W_F - mgh\right)}{m}}$

Put in the numbers: $v_2 = \sqrt{\frac{2(11.25 + 500.0 - 184.5 - 167.76)}{10.0}}$

$v_2 = \sqrt{\frac{2(158.99)}{10.0}} = 5.6390\ \text{m/s}$

FINAL ANSWER

(a) $W_g = -168$ J

(b) Energy lost to friction = 185 J

(c) $W_{app} = 5.00 \times 10^2$ J

(d) $v_2 = 5.64$ m/s

Worksheet answer: (b) 184 J; (d) 5.65 m/s
Why the answers differ: Different rounding. (b) (36.9)(5.00) = 184.5 J, which rounds up to 185 J; the worksheet cut it off at 184 J. (d) With the unrounded works the speed is 5.6390 m/s, which rounds to 5.64 m/s. The key could not get 5.65 m/s from the given data.

QUESTION 7

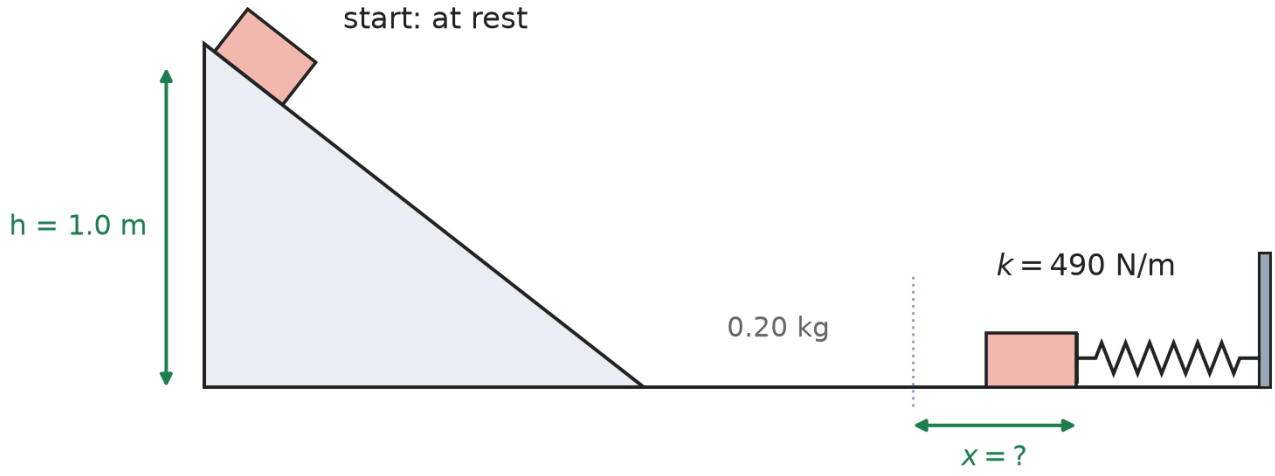
Block down a track onto a spring

7. A block of mass 0.20 kg is released from rest and slides down a frictionless track of that is 1.0 m above a table. At the bottom of the track, where the surface is horizontal, the block strikes and sticks to a light spring with a force constant of 490 N/m. Find the maximum distance the spring is compressed [For an extra challenge, solve this problem in terms of m, k and h].

GIVEN

$m = 0.20$ kg $\quad h = 1.0$ m $\quad k = 490$ N/m $\quad x = ?$

The block at release and at maximum compression (h = 0 on the table)



Start: at rest at height h, so the energy is all gravitational. End: at rest against the fully compressed spring, so it is all elastic. The track is frictionless. Solving in letters first also answers the extra challenge.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mgh = \frac{1}{2}kx^2$

Multiply by $\frac{2}{k}$ and square root: $x = \sqrt{\frac{2mgh}{k}}$

Put in the numbers: $x = \sqrt{\frac{2(0.20)(9.81)(1.0)}{490}} = 0.08949 \text{ m}$

FINAL ANSWER

x = 0.089 m = 8.9 cm

In terms of m, k and h: x = √(2mgh/k)

QUESTION 8

Mass, pulley and spring (Fig 1)

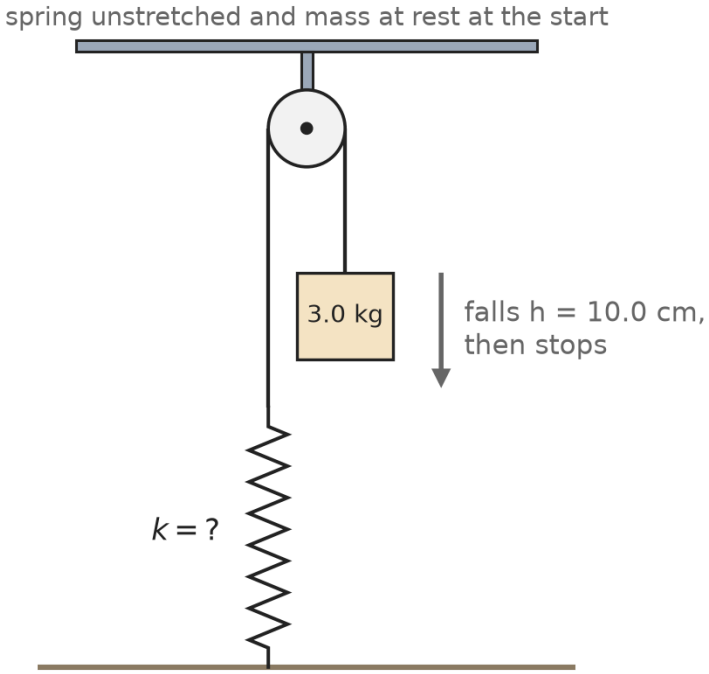
8. A 3.0 kg mass is fastened to a light spring that passes over a pulley as shown (Fig 1). The pulley is frictionless and the mass is released from rest when the spring is unstretched. If the mass drops a distance of 10.0 cm before coming to rest, find the force constant of the spring.

GIVEN

$m = 3.0 \text{ kg}$ $h = 10.0 \text{ cm} = 0.100 \text{ m}$

$k = ?$

Fig 1: the mass, the pulley and the spring



When the mass drops 10.0 cm the string pulls the top of the spring up 10.0 cm, so the spring stretches x = h.

Take h = 0 at the lowest point of the mass. Start: at rest, 0.100 m above h = 0, spring unstretched. End: at rest at h = 0 with the spring stretched x = h.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mgh = \frac{1}{2}kh^2$

Divide by h, then multiply by $\frac{2}{h}$: $k = \frac{2mg}{h}$

Put in the numbers: $k = \frac{2(3.0)(9.81)}{0.100} = 588.6 \text{ N/m}$

FINAL ANSWER

k = 5.9 × 10^2 N/m

Worksheet answer: 589 N/m
Why the answers differ: Same value, different rounding. 2(3.0)(9.81)/0.100 = 588.6 N/m. The mass (3.0 kg) has 2 significant figures, so the key gives 5.9 × 10^2 N/m; the worksheet keeps 3.

QUESTION 9

Spring toy (Fig 2)

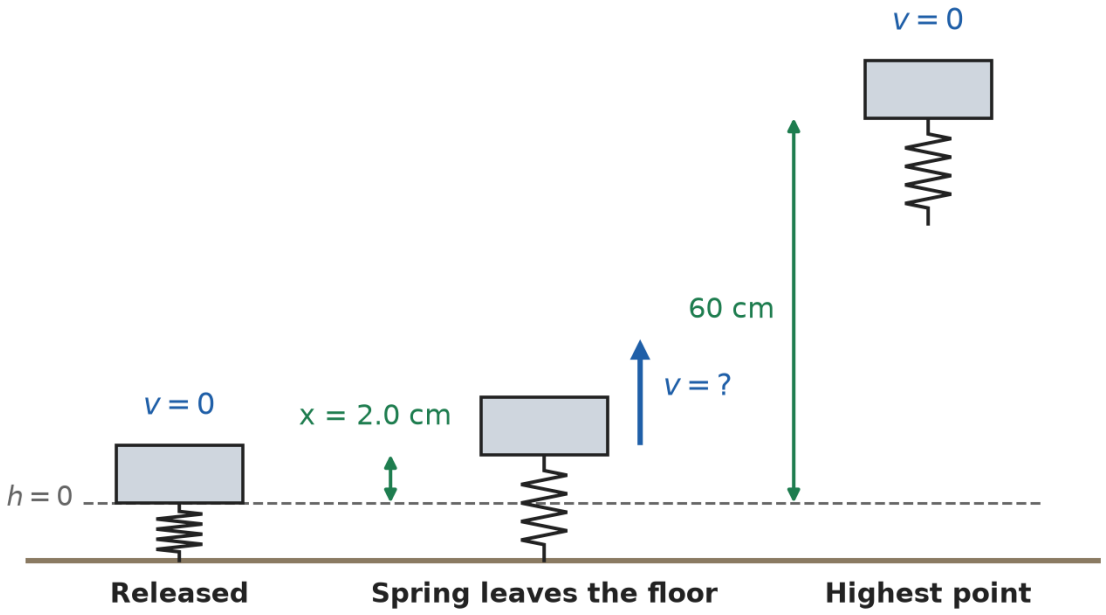
9. A child's toy of mass 30.0 g consists of a piece of plastic attached to a spring as shown (Fig 2). The spring is compressed against the floor a distance of 2.0 cm and the toy is released. The maximum height the toy reaches is 60 cm. (a) Determine the force constant of the spring. (b) How fast is the toy going the instant the spring leaves the ground?

GIVEN

$m = 30.0\ \text{g} = 0.0300\ \text{kg}$ $x = 2.0\ \text{cm} = 0.020\ \text{m}$

$h = 60\ \text{cm} = 0.60\ \text{m}$ $k = ?$

The toy at release, as the spring leaves the floor, and at its highest point



h = 0 is the release position. The 60 cm is measured up from there.

(a) Determine the force constant of the spring.

Start: at rest on the compressed spring, so the energy is all elastic. Highest point: at rest, spring unstretched, so it is all gravitational.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}kx^2 = mgh$

Multiply by $\frac{2}{x^2}$: $k = \frac{2mgh}{x^2}$

Put in the numbers: $k = \frac{2(0.0300)(9.81)(0.60)}{(0.020)^2} = 882.9 \text{ N/m}$

(b) How fast is the toy going the instant the spring leaves the ground?

The spring leaves the floor when it is back to its unstretched length. The toy has then risen x = 0.020 m and is moving, so it has E_g and E_k.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}kx^2 = mgx + \frac{1}{2}mv^2$

Subtract mgx, multiply by $\frac{2}{m}$, square root: $v = \sqrt{\frac{kx^2}{m} - 2gx}$

Put in the numbers: $v = \sqrt{\frac{(882.9)(0.020)^2}{0.0300} - 2(9.81)(0.020)}$

$v = \sqrt{11.7720 - 0.3924} = 3.373 \text{ m/s}$

FINAL ANSWER

(a) $k = 8.8 \times 10^2$ N/m

(b) v = 3.4 m/s

Worksheet answer: 883 N/m, 3.4 m/s
Why the answers differ: (a) Same value, different rounding. $2(0.0300)(9.81)(0.60)/(0.020)^2 = 882.9$ N/m. The compression (2.0 cm) and the height (60 cm) have 2 significant figures, so the key gives 8.8×10^2 N/m; the worksheet keeps 3.

QUESTION 10

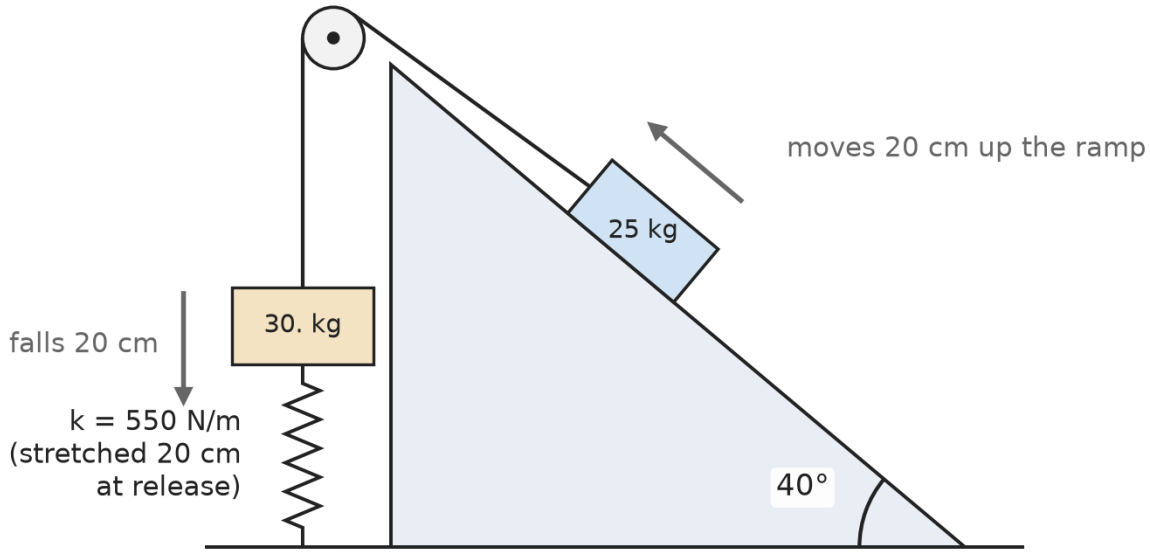
Two blocks, a ramp and a spring (Fig. 3)

10. Two blocks are attached and place on an incline as shown (Fig. 3). The 30 kg mass is attached to the spring (k = 550 N/m). The pulley and ramp are both frictionless and in the set-up shown, the spring is unstretched. The 25 kg block is pulled 20 cm down the ramp so that the 30 kg block is now 40 cm above the floor and the spring is stretched. The system is then released. Determine the speed of both blocks when the spring returns to equilibrium.

GIVEN

$m_{30} = 30.\ \text{kg}$	$m_{25} = 25\ \text{kg}$	$k = 550\ \text{N/m}$
$x = 20\ \text{cm} = 0.20\ \text{m}$	$\theta = 40°$	$v = ?$

Fig. 3 at the moment of release



The stretched spring and gravity both pull the 30. kg block down, so it falls 20 cm back to where the spring is unstretched, and the 25 kg block is pulled 20 cm up the ramp. The string does not stretch, so both blocks have the same speed v. Each block gets its own h = 0 at its lowest point.

Step 1 – Energy equation

Start (at rest): the spring is stretched x, and the 30. kg block is x above its lowest point. End: both blocks are moving, the spring is unstretched, and the 25 kg block has risen x sin θ.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies:

$$\frac{1}{2}kx^2 + m_{30}gx = m_{25}g(x\sin\theta) + \frac{1}{2}(m_{30} + m_{25})v^2$$

Isolate v: $$v = \sqrt{\frac{2\left(\frac{1}{2}kx^2 + m_{30}gx - m_{25}gx\sin\theta\right)}{m_{30} + m_{25}}}$$

Step 2 – Put in the numbers

Spring: $\frac{1}{2}kx^2 = \frac{1}{2}(550)(0.20)^2 = 11.00$ J

30. kg block: $m_{30}gx = (30.)(9.81)(0.20) = 58.86$ J

25 kg block: $m_{25}gx\sin\theta = (25)(9.81)(0.20)\sin 40° = 31.53$ J

Speed: $$v = \sqrt{\frac{2(11.00 + 58.86 - 31.53)}{30. + 25}} = 1.181 \text{ m/s}$$

FINAL ANSWER

v = 1.2 m/s for both blocks

QUESTION 11

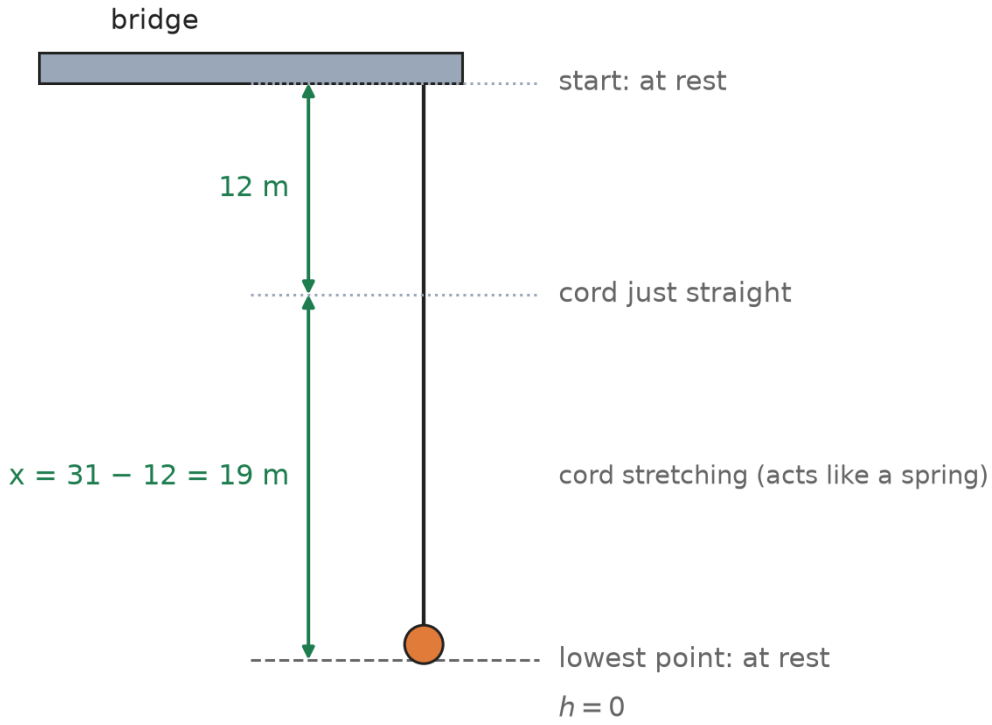
Bungee jumper

11. A 60-kg bungee jumper jumps from a bridge. She is tied to a 12-m-long bungee cord and falls a total of 31 m. (*a*) Calculate the spring constant *k* of the bungee cord. (*b*) Calculate the maximum acceleration experienced by the jumper.

GIVEN

$m = 60$ kg	cord length $= 12$ m	total fall $= 31$ m
$k = ?$	$a = ?$	

The jump (h = 0 at the lowest point)



The cord is slack for the first 12 m. It only stretches during the last 31 − 12 = 19 m.

(a) Calculate the spring constant k of the bungee cord.

Start: at rest on the bridge, 31 m above h = 0, so the energy is all gravitational. End: at rest at the lowest point with the cord stretched x = 19 m, so it is all elastic.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mgh = \frac{1}{2}kx^2$

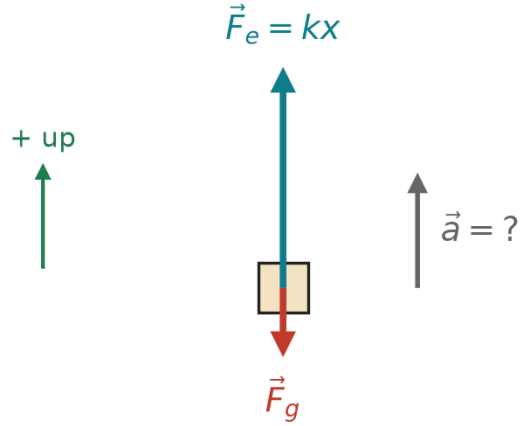
Multiply by $\frac{2}{x^2}$: $k = \frac{2mgh}{x^2}$

Put in the numbers: $k = \frac{2(60)(9.81)(31)}{(19)^2} = 101.09 \text{ N/m}$

(b) Calculate the maximum acceleration experienced by the jumper.

FBD of the jumper at the lowest point (+y up, the direction of the acceleration)

Jumper at the lowest point



The cord force is largest when the cord is stretched the most, at the lowest point. There it is bigger than gravity, so the acceleration is up. Forces are sizes, with + up:

Formula: $F_{net} = ma$

Put in the forces: $kx - mg = ma$

Divide both sides by m: $a = \dfrac{kx - mg}{m}$

Put in the numbers: $a = \dfrac{(101.09)(19) - (60)(9.81)}{60}$

$$a = \frac{1920.7 - 588.6}{60} = +22.20 \text{ m/s}^2$$

FINAL ANSWER

(a) k = 1.0 × 10^2 N/m

(b) a = +22 m/s^2

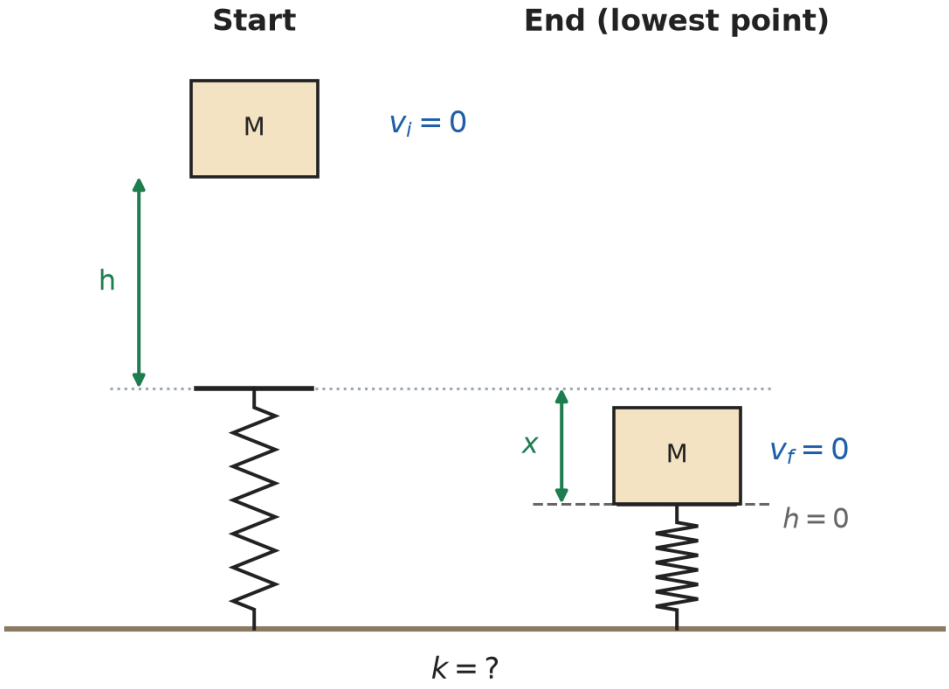
$\vec{a}$ = 22 m/s^2 [up]

QUESTION 12

Designing the elevator spring

12. An engineer is designing a spring to be placed at the bottom of an elevator shaft. If the elevator cable should happen to break at a height of h above the top of the spring, calculate the value that the spring constant k should have so that passengers undergo an acceleration of no more than 5.0 g when brought to rest. Let M be the total mass of the elevator and passengers.

The elevator when the cable breaks and at its lowest point



x is the maximum compression of the spring. h = 0 is the lowest point.

Step 1 – Largest acceleration (at the lowest point)

The spring force is largest at maximum compression, so the acceleration is largest there, and it points up. Set it equal to 5.0g. Forces are sizes, with + up:

Formula: $F_{net} = Ma$

Put in the forces and $a = 5.0g$: $kx - Mg = M(5.0g)$

Add Mg, divide by k: $x = \dfrac{6.0Mg}{k}$

Step 2 – Energy equation

Start: at rest, (h + x) above the lowest point. End: at rest on the fully compressed spring.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $Mg(h+x) = \frac{1}{2}kx^2$

Step 3 – Put Step 1 into Step 2 and isolate k

Put in $x = \frac{6.0Mg}{k}$: $Mgh + Mg\left(\frac{6.0Mg}{k}\right) = \frac{1}{2}k\left(\frac{6.0Mg}{k}\right)^2$

Simplify each term: $Mgh + \frac{6.0M^2g^2}{k} = \frac{18M^2g^2}{k}$

Subtract $\frac{6.0M^2g^2}{k}$: $Mgh = \frac{12M^2g^2}{k}$

Multiply by k, divide by Mgh: $k = \frac{12Mg}{h}$

A stiffer spring would stop the elevator in a shorter distance with a larger acceleration, so this is the largest k allowed.

FINAL ANSWER

k = 12Mg/h (or less)

QUESTION 13

Block colliding with a spring

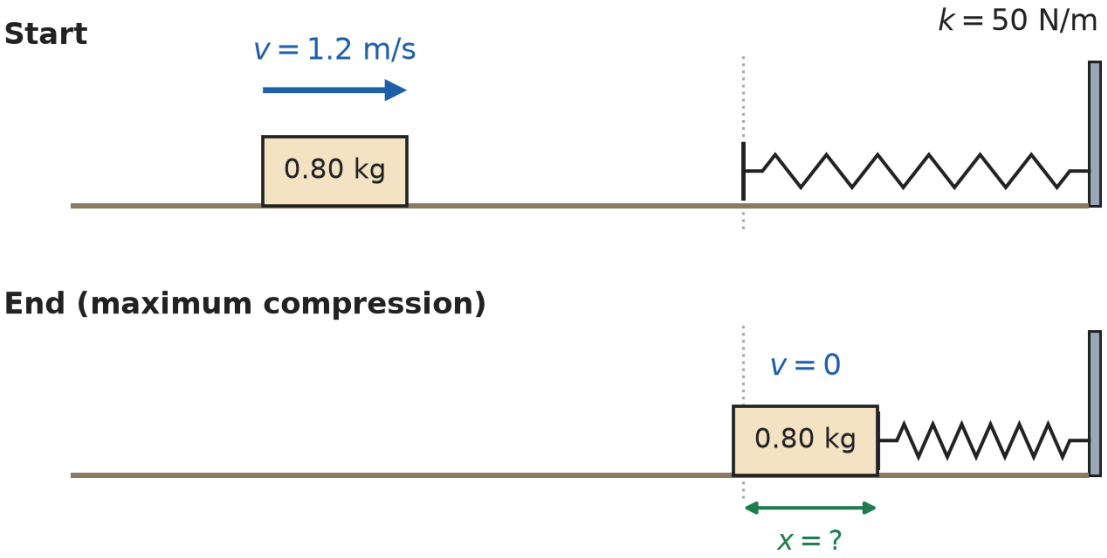
13. A block of mass 0.80 kg is given an initial velocity of 1.2 m/s to the right and collides with a spring of negligible mass of force constant 50 N/m. (a) Assuming the surface to be frictionless, what is the maximum compression of the spring? (b) If the surface has a coefficient of kinetic friction of 0.50, what would be the maximum compression?

GIVEN

$m = 0.80$ kg $\quad v = 1.2$ m/s $\quad k = 50$ N/m

$\mu_k = 0.50$ (part b) $\quad x = ?$

The block as it reaches the spring and at maximum compression



(a) Assuming the surface to be frictionless, what is the maximum compression of the spring?

Start: all kinetic. End: at rest, all elastic. The surface is level, so E_g does not change.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$

Multiply by $\frac{2}{k}$ and square root: $x = v\sqrt{\frac{m}{k}}$

Put in the numbers: $x = (1.2)\sqrt{\frac{0.80}{50}} = 0.1518 \text{ m}$

(b) If the surface has a coefficient of kinetic friction of 0.50, what would be the maximum compression?

Step 1 – Friction force (vertical balance first)

The block does not accelerate vertically. Friction acts while the block slides the distance x.

Balance: $F_{net,y} = 0, \quad \text{so} \quad 0 = F_N - F_g, \quad \text{so} \quad F_N = mg$

Friction: $F_f = \mu_k F_N = \mu_k mg = (0.50)(0.80)(9.81) = 3.924 \text{ N}$

Step 2 – Energy equation with friction

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies: $\frac{1}{2}mv^2 - F_f x = \frac{1}{2}kx^2$

Move everything to one side: $\frac{1}{2}kx^2 + F_f x - \frac{1}{2}mv^2 = 0$

Put in the numbers: $25.0x^2 + 3.924x - 0.576 = 0$

Step 3 – Solve the quadratic

Quadratic formula: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Put in the numbers:

$$x = \frac{-(3.924) \pm \sqrt{(3.924)^2 - 4(25.0)(-0.576)}}{2(25.0)}$$

$$x = \frac{-3.924 \pm 8.5439}{50.0}$$

$x = 0.0924$ m or $x = -0.2494$ m

A compression cannot be negative, so x = 0.09240 m.

FINAL ANSWER

(a) x = 0.15 m

(b) x = 0.092 m

QUESTION 14

Two masses, a spring and friction

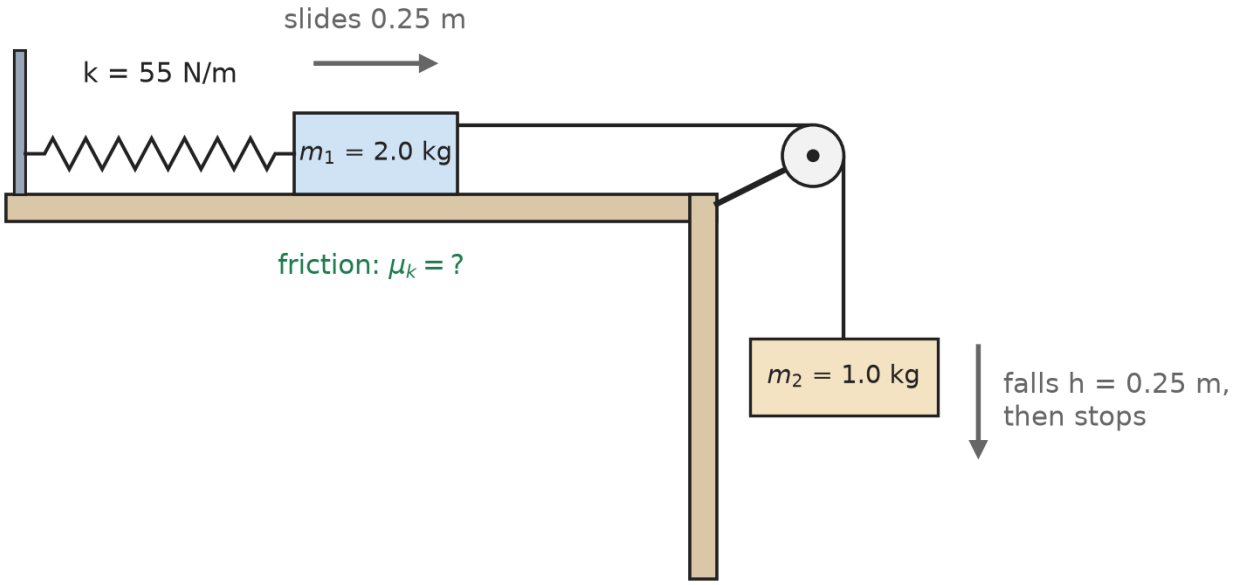
14. Two masses (m_1 = 2.0 kg and m_2 = 1.0 kg) are connected by a light string that passes over a frictionless pulley as shown. The system is released from rest when the spring (k = 55 N/m) is unstretched. If the hanging mass falls a distance h=0.25 m before coming to rest, calculate the co-efficient of kinetic friction between the mass on the desk and the surface. For an extra challenge, find the co-efficient in terms of m_1, m_2, k, h and g.

GIVEN

$m_1 = 2.0$ kg $\quad m_2 = 1.0$ kg $\quad k = 55$ N/m

$h = 0.25$ m $\quad \mu_k = ?$

The system at the moment of release



The string does not stretch: when m_2 falls h, m_1 slides h and the spring stretches h.

Step 1 – Friction force on m_1 (vertical balance first)

Balance: $F_{net,y} = 0$, so $0 = F_N - m_1 g$, so $F_N = m_1 g$

Friction: $F_f = \mu_k F_N = \mu_k m_1 g$

Work removed over the distance h: $W_F = \mu_k m_1 g h$

Step 2 – Energy equation with friction

Take h = 0 at the lowest point of m_2. Start: at rest, m_2 is h above its lowest point, spring unstretched. End: at rest, spring stretched h. m_1 stays at the same height.

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies: $m_2gh - \mu_k m_1 gh = \frac{1}{2}kh^2$

Divide both sides by h: $m_2g - \mu_k m_1 g = \frac{1}{2}kh$

Isolate μ_k: $\mu_k = \frac{m_2g - \frac{1}{2}kh}{m_1 g}$

Put in the numbers:

$$\mu_k = \frac{(1.0)(9.81) - \frac{1}{2}(55)(0.25)}{(2.0)(9.81)} = \frac{9.81 - 6.875}{19.62} = 0.1496$$

FINAL ANSWER

μ_k = 0.15

In letters: $\mu_k = (m_2g - \frac{1}{2}kh) / (m_1g)$

QUESTION 15

Block on a spring with friction

15. A 200 g wood block is firmly attached to a horizontal spring. The block can slide along a table where the coefficient of friction is 0.40. A force of 20 N compresses the spring 18 cm. If the spring is released from this position, how far beyond its equilibrium position will it stretch on its first swing?

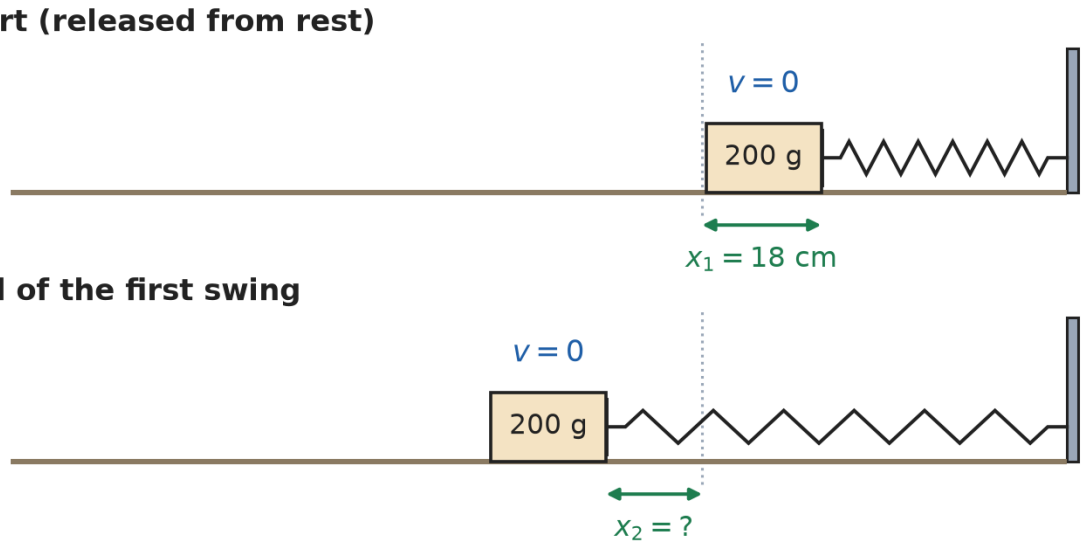
GIVEN

$m = 200\ \text{g} = 0.200\ \text{kg}$ $\quad \mu = 0.40$

$F = 20\ \text{N}$ $\quad x_1 = 18\ \text{cm} = 0.18\ \text{m}$

$x_2 = ?$

The block at release and at the end of its first swing



the block is attached to the spring; it slides a total distance $x_1 + x_2$ against friction

Step 1 – Spring constant

A 20 N force holds the spring compressed 18 cm.

Hooke's law: $F = kx_1 \quad \Rightarrow \quad k = \dfrac{F}{x_1}$

Put in the numbers: $k = \dfrac{20}{0.18} = 111.11\ \text{N/m}$

Step 2 – Friction force (vertical balance first)

Balance: $F_{net,y} = 0, \quad \text{so} \quad 0 = F_N - F_g, \quad \text{so} \quad F_N = mg$

Friction: $F_f = \mu F_N = \mu mg = (0.40)(0.200)(9.81) = 0.7848 \text{ N}$

Step 3 – Energy equation with friction

Start: at rest, spring compressed x_1. End of the swing: at rest, spring stretched x_2. The block slides $x_1 + x_2$ in between, so friction removes $F_f(x_1 + x_2)$.

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies: $\frac{1}{2}kx_1^2 - F_f(x_1 + x_2) = \frac{1}{2}kx_2^2$

Collect the spring terms: $\frac{1}{2}k\left(x_1^2 - x_2^2\right) = F_f(x_1 + x_2)$

Factor the difference of squares:

$$\frac{1}{2}k(x_1 - x_2)(x_1 + x_2) = F_f(x_1 + x_2)$$

Divide both sides by $(x_1 + x_2)$: $\frac{1}{2}k(x_1 - x_2) = F_f$

Isolate x_2: $x_2 = x_1 - \frac{2F_f}{k}$

Put in the numbers:

$$x_2 = 0.18 - \frac{2(0.7848)}{111.11} = 0.18 - 0.01413 = 0.16587 \text{ m}$$

FINAL ANSWER

x_2 = 0.17 m beyond the equilibrium position

Worksheet answer: 0.166 m
Why the answers differ: Same value, different rounding. 0.18 − 0.01413 = 0.16587 m. The data (200 g, 0.40, 20 N, 18 cm) have 2 significant figures, so the key gives 0.17 m; the worksheet keeps 3.

QUESTION 16

Out and back with friction

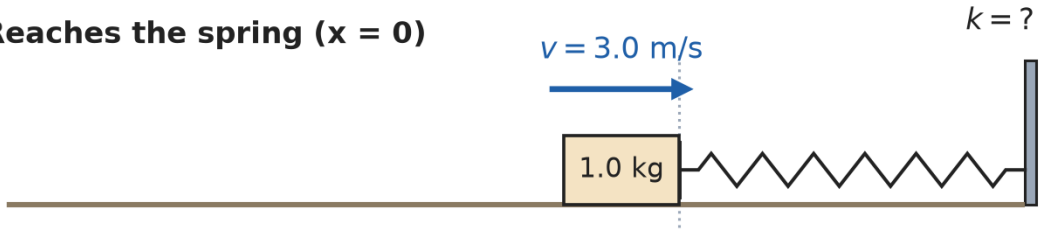
16. An object of mass m = 1.0 kg is travelling on a horizontal surface. There is a co-efficient of kinetic friction, μ = 0.20, between the object and the surface. The object has speed v = 3.0 m/s when it encounters a spring. It compresses the spring, stops and then recoils and travels in the opposite direction. When it returns to the point at which it encountered the spring (i.e. x = 0), it stops. Find the spring constant, k. For an extra challenge, find k in terms of the mass, initial speed and co-efficient of friction.

GIVEN

$m = 1.0$ kg $\quad \mu = 0.20$ $\quad v = 3.0$ m/s $\quad k = ?$

The three moments in the motion

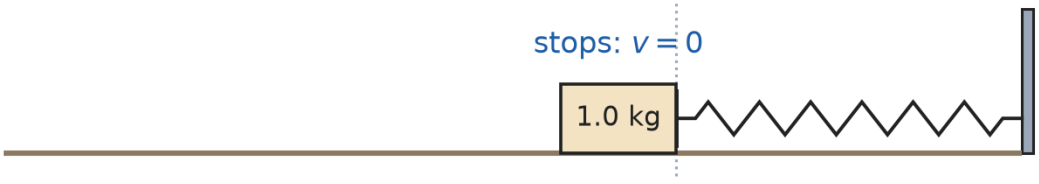
1. Reaches the spring (x = 0)



2. Maximum compression



3. Back at x = 0



Step 1 – Friction force (vertical balance first)

Balance: $F_{net,y} = 0$, so $0 = F_N - F_g$, so $F_N = mg$

Friction: $F_f = \mu F_N = \mu mg$

Step 2 – Energy equation for each half of the trip

Call the maximum compression x. Friction removes μmgx on the way in and another μmgx on the way out.

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Way in (moment 1 to 2): $\frac{1}{2}mv^2 - \mu mgx = \frac{1}{2}kx^2$

Way out (moment 2 to 3): $\frac{1}{2}kx^2 - \mu mgx = 0$

Step 3 – Find the compression x

The way-out equation says ½kx^2 = μmgx. Put that into the way-in equation.

Put in $\frac{1}{2}kx^2 = \mu mgx$: $\frac{1}{2}mv^2 - \mu mgx = \mu mgx$

Add μmgx, divide by $2\mu mg$: $x = \frac{v^2}{4\mu g}$

Put in the numbers: $x = \frac{(3.0)^2}{4(0.20)(9.81)} = 1.1468 \text{ m}$

Step 4 – Find k

Use the way-out equation again and divide it by x.

Way out: $\frac{1}{2}kx = \mu mg \quad \Rightarrow \quad k = \frac{2\mu mg}{x}$

Put in $x = \frac{v^2}{4\mu g}$: $k = \frac{8\mu^2 m g^2}{v^2}$

Put in the numbers: $k = \frac{8(0.20)^2(1.0)(9.81)^2}{(3.0)^2} = 3.4217 \text{ N/m}$

FINAL ANSWER

k = 3.4 N/m

In letters: $k = 8\mu^2 m g^2 / v^2$

QUESTION 17

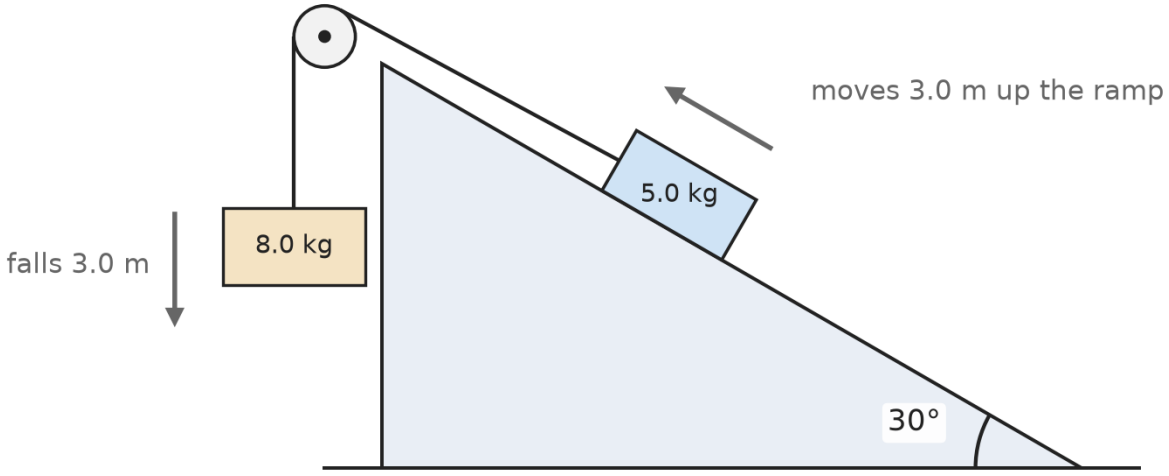
Two blocks and a ramp

17. Two blocks are attached via a pulley as shown below. If the hanging mass is released and allowed to fall, how fast will it be going after it has fallen 3.0 m (a) assuming there is no friction (b) assuming there are 10. N of friction opposing the motion.

GIVEN

$m_h = 8.0$ kg	$m_r = 5.0$ kg	$\theta = 30°$
$\Delta d = 3.0$ m	$F_f = 10.$ N (part b)	$v = ?$

The two blocks (hanging mass falls, ramp block moves up the ramp)



The string does not stretch, so the ramp block moves 3.0 m up the ramp and both blocks have the same speed v. The ramp block rises Δd sin θ. Each block gets its own h = 0 at its lowest point, and the system starts from rest.

(a) assuming there is no friction

Start: the hanging mass is Δd above its lowest point. End: the ramp block has risen Δd sin θ and both blocks are moving.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $m_h g\,\Delta d = m_r g(\Delta d \sin\theta) + \frac{1}{2}(m_h + m_r)v^2$

Isolate v: $v = \sqrt{\frac{2(m_h g\,\Delta d - m_r g\,\Delta d \sin\theta)}{m_h + m_r}}$

Hanging mass: $m_h g\,\Delta d = (8.0)(9.81)(3.0) = 235.44\ \text{J}$

Ramp block: $m_r g\,\Delta d \sin\theta = (5.0)(9.81)(3.0)\sin 30° = 73.575\ \text{J}$

Speed: $v = \sqrt{\frac{2(235.44 - 73.575)}{8.0 + 5.0}} = 4.990\ \text{m/s}$

(b) assuming there are 10. N of friction opposing the motion.

Friction acts over the 3.0 m, so it removes $W_F = F_f \Delta d = (10.)(3.0) = 30.0$ J.

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies:

$$m_h g\,\Delta d - F_f \Delta d = m_r g(\Delta d \sin\theta) + \frac{1}{2}(m_h + m_r)v^2$$

Isolate v: $$v = \sqrt{\frac{2(m_h g\,\Delta d - F_f \Delta d - m_r g\,\Delta d \sin\theta)}{m_h + m_r}}$$

Put in the numbers:

$$v = \sqrt{\frac{2(235.44 - 30.0 - 73.575)}{8.0 + 5.0}} = 4.504 \text{ m/s}$$

FINAL ANSWER

(a) v = 5.0 m/s

(b) v = 4.5 m/s

QUESTION 18

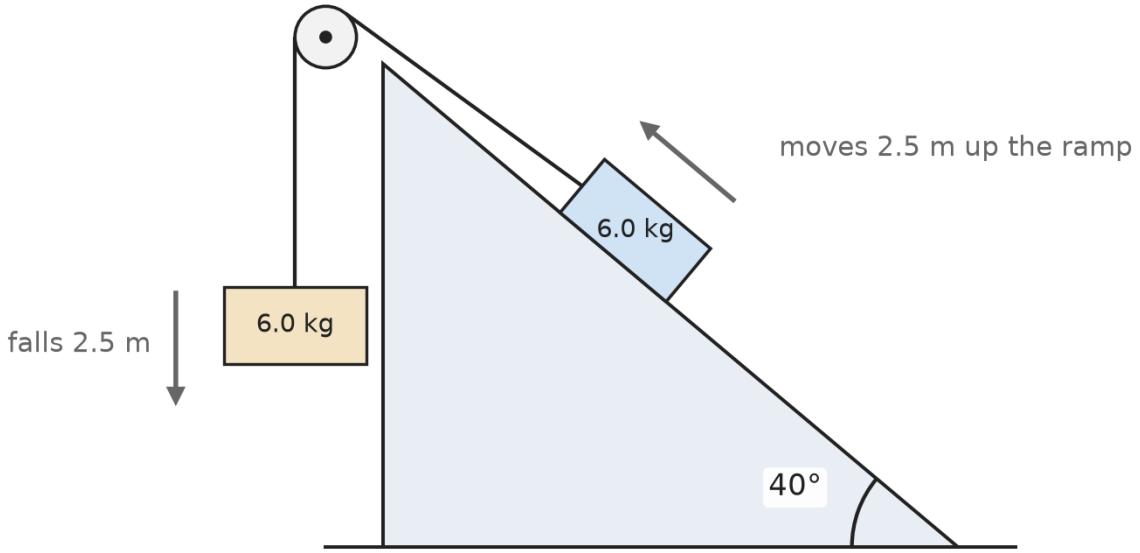
Finding the friction force

18. Two blocks are attached via a pulley as shown below. If the hanging mass is released and allowed to fall, determine the force of friction if the blocks are moving 2.0 m/s after the one on the ramp has moved up 2.5 m.

GIVEN

$m_h = 6.0$ kg	$m_r = 6.0$ kg	$\theta = 40°$
$\Delta d = 2.5$ m	$v = 2.0$ m/s	$F_f = ?$

The two blocks (hanging mass falls, ramp block moves up the ramp)



Step 1 – Energy equation with friction

The system starts from rest. The hanging mass falls 2.5 m, the ramp block rises Δd sin θ, and both end with speed v. Friction acts on the ramp block over the 2.5 m.

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies:

$$m_h g\,\Delta d - F_f \Delta d = m_r g(\Delta d \sin\theta) + \frac{1}{2}(m_h + m_r)v^2$$

Isolate F_f: $$F_f = \frac{m_h g\,\Delta d - m_r g\,\Delta d \sin\theta - \frac{1}{2}(m_h + m_r)v^2}{\Delta d}$$

Step 2 – Put in the numbers

Hanging mass: $m_h g\,\Delta d = (6.0)(9.81)(2.5) = 147.15\ \text{J}$

Ramp block: $m_r g\,\Delta d \sin\theta = (6.0)(9.81)(2.5)\sin 40° = 94.59\ \text{J}$

Kinetic energy: $\frac{1}{2}(m_h + m_r)v^2 = \frac{1}{2}(6.0 + 6.0)(2.0)^2 = 24.0\ \text{J}$

Friction: $F_f = \frac{147.15 - 94.59 - 24.0}{2.5} = 11.43\ \text{N}$

FINAL ANSWER

F_f = 11 N (down the ramp, against the motion)

QUESTION 19

Pinball launcher

19. The ball launcher in a pinball machine has a spring that has a force constant of 1.20 N/cm. The surface on which the ball moves is inclined 10.0^{o} with respect to the horizontal. If the spring is initially compressed 5.00 cm, find the launching speed of a 100 g ball when the spring is released. Friction and the mass of the spring are negligible.

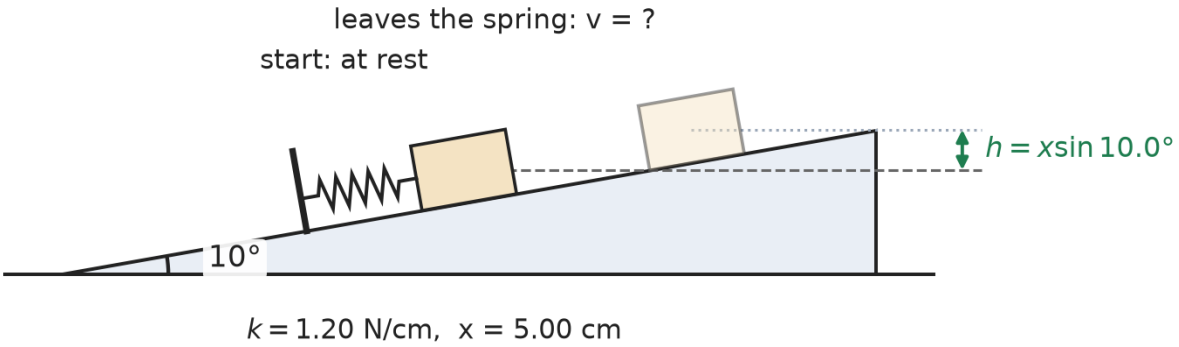
GIVEN

$k = 1.20\ \text{N/cm} = 120.\ \text{N/m}$ $\theta = 10.0°$

$x = 5.00\ \text{cm} = 0.0500\ \text{m}$ $m = 100\ \text{g} = 0.100\ \text{kg}$

$v = ?$

The ball on the compressed spring and as it leaves the spring



The ball leaves the spring when the spring is back to its unstretched length, after moving x = 5.00 cm up the incline. It has then risen h = x sin θ.

Step 1 – Energy equation

Take h = 0 at the start. Start: at rest, all elastic. As the ball leaves the spring: kinetic plus a little gravitational.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + mg(x\sin\theta)$

Isolate v: $v = \sqrt{\frac{2\left(\frac{1}{2}kx^2 - mgx\sin\theta\right)}{m}}$

Step 2 – Put in the numbers

Spring: $\frac{1}{2}kx^2 = \frac{1}{2}(120.)(0.0500)^2 = 0.15000\ \text{J}$

Gravity:

$mgx\sin\theta = (0.100)(9.81)(0.0500)\sin 10.0° = 0.008517\ \text{J}$

Speed: $v = \sqrt{\frac{2(0.15000 - 0.008517)}{0.100}} = 1.6822\ \text{m/s}$

FINAL ANSWER

v = 1.68 m/s

QUESTION 20

Pulley arrangement

20. A block of mass m = 2.0 kg hangs on the end of a cord and is connected to a block of mass M = 6.0 kg by the pulley arrangement shown. Using energy considerations only, (a) find the speed of m when it has fallen 20. cm assuming the blocks are at rest and there is no friction (b) Repeat (a) assuming there is a co-efficient of kinetic friction, μ = 0.15 , between M and the table.

GIVEN

$m = 2.0$ kg

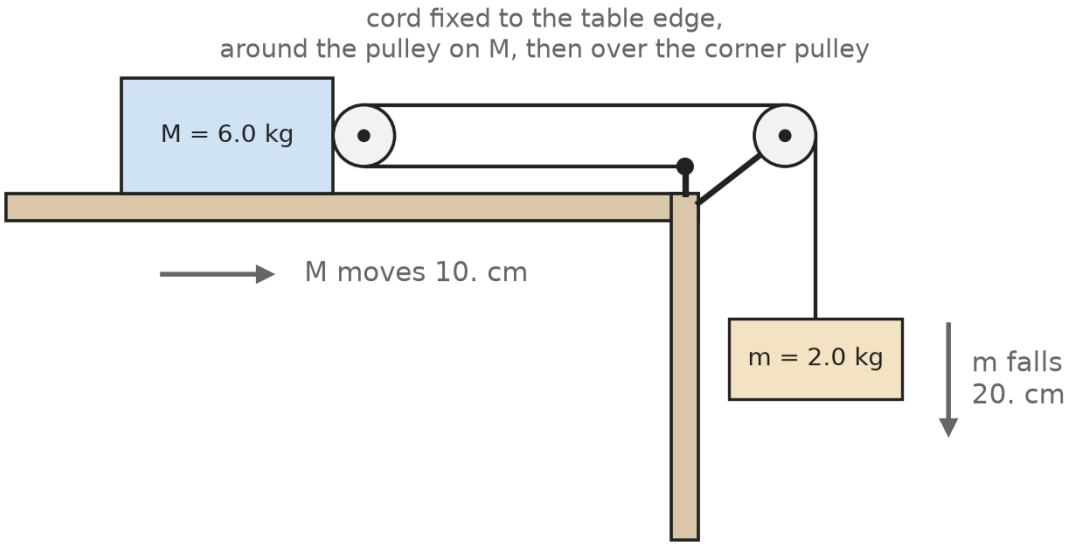
$M = 6.0$ kg

$h = 20.\ \text{cm} = 0.20$ m

$\mu = 0.15$ (part b)

$v = ?$

The pulley arrangement



When M moves a distance d toward the corner, both pieces of cord between M and the corner get shorter by d, so m falls 2d.

In the picture the cord is fixed to the edge of the table, goes around a pulley attached to M, and comes back over the corner pulley to m. So M moves half as far as m and, at every moment, half as fast: when m has speed v, M has speed v/2. M stays at the same height, so only m changes its E_g.

(a) find the speed of m when it has fallen 20. cm assuming the blocks are at rest and there is no friction

Take h = 0 at the lowest point of m. Start: at rest, m is h above it. End: m moves at v and M moves at v/2.

Formula: $E_{TOTi} = E_{TOTf}$

Put in the energies: $mgh = \frac{1}{2}mv^2 + \frac{1}{2}M\left(\frac{v}{2}\right)^2$

Factor out $\frac{1}{2}v^2$: $mgh = \frac{1}{2}v^2\left(m + \frac{M}{4}\right)$

Isolate v: $v = \sqrt{\frac{2mgh}{m + \frac{M}{4}}}$

Put in the numbers:

$$v = \sqrt{\frac{2(2.0)(9.81)(0.20)}{2.0 + \frac{6.0}{4}}} = \sqrt{\frac{7.848}{3.5}} = 1.497 \text{ m/s}$$

(b) Repeat (a) assuming there is a co-efficient of kinetic friction, μ = 0.15 , between M and the table.

Step 1 – Work removed by friction

M does not accelerate vertically. M slides only h/2 = 0.10 m while m falls 0.20 m.

Balance: $F_{net,y} = 0, \quad \text{so} \quad 0 = F_N - Mg, \quad \text{so} \quad F_N = Mg$

Friction: $F_f = \mu F_N = \mu Mg = (0.15)(6.0)(9.81) = 8.829 \text{ N}$

Work removed: $W_F = F_f\left(\frac{h}{2}\right) = (8.829)(0.10) = 0.8829 \text{ J}$

Step 2 – Energy equation with friction

Formula: $E_{TOTi} - W_F = E_{TOTf}$

Put in the energies: $mgh - W_F = \frac{1}{2}v^2\left(m + \frac{M}{4}\right)$

Isolate v: $v = \sqrt{\frac{2(mgh - W_F)}{m + \frac{M}{4}}}$

Put in the numbers: $v = \sqrt{\frac{2(3.924 - 0.8829)}{3.5}} = 1.318 \text{ m/s}$

FINAL ANSWER

(a) v = 1.5 m/s

(b) v = 1.3 m/s